# 1 Threshold Sweep

Table 1: Threshold-sensitivity analysis across two CARLA simulation campaigns. A scenario is counted as successful if the calibrated ensemble mean reaches threshold $\theta_{\mathcal{S}}$ before the stopping-distance boundary. Success rates are reported as percentages with $95\%$ bootstrap confidence intervals in brackets. Distance-to-threshold is reported as median [IQR] for scenarios in which the threshold is reached. A dash indicates that no scenario reached the corresponding threshold.

| $\theta_{\mathcal{S}}$ | Experiment 1, $N = 972$ | | Experiment 2, $N = 1054$ | |
|---|---|---|---|---|
| | Success before $sd$ (%) | $d_{\mathcal{S}}$ (m) | Success before $sd$ (%) | $d_{\mathcal{S}}$ (m) |
| 0.60 | 30.9 [28.1, 33.6] | 22.55 [16.55, 25.55] | 31.4 [28.6, 34.3] | 19.55 [16.55, 25.55] |
| 0.65 | 24.3 [21.6, 26.9] | 19.55 [16.55, 22.55] | 26.8 [24.1, 29.4] | 19.55 [16.55, 25.55] |
| 0.70 | 21.5 [18.9, 24.1] | 19.55 [13.55, 22.55] | 22.2 [19.6, 24.8] | 19.55 [13.55, 22.55] |
| 0.75 | 19.2 [16.9, 21.6] | 19.55 [13.55, 22.55] | 18.5 [16.2, 20.8] | 16.55 [13.55, 22.55] |
| 0.80 | 15.7 [13.6, 18.0] | 16.55 [13.55, 22.55] | 12.9 [10.8, 15.1] | 16.55 [10.55, 19.55] |
| 0.85 | 0.9 [0.4, 1.5] | 16.55 [10.55, 19.55] | 3.8 [2.7, 4.9] | 13.55 [10.55, 19.55] |
| 0.90 | 0.0 [0.0, 0.0] | 10.55 [7.55, 13.55] | 0.0 [0.0, 0.0] | 10.55 [7.55, 13.55] |
| 0.95 | 0.0 [0.0, 0.0] | 4.55 [4.55, 4.55] | 0.0 [0.0, 0.0] | 4.55 [4.55, 4.55] |
| 1.00 | 0.0 [0.0, 0.0] | – | 0.0 [0.0, 0.0] | – |

## 2 Effect of Fog Density, Cloudiness and Precipitation

Table 2: Fog-density sensitivity with cloudiness and precipitation fixed to zero, reported separately for adversary-inactive and adversary-active scenarios. A scenario is counted as successful if the calibrated ensemble mean reaches $\theta_{\mathcal{S}} = 0.75$ before the stopping-distance boundary. Detection-before-stopping values are reported as percentages with 95% bootstrap confidence intervals in brackets. $d_{\mathcal{S}}$ is reported as median [IQR]. Critical-zone FNR is computed over frames whose distance to the stop sign lies in $[sd, sd + \Delta]$, where $\Delta = 10$ m.

| Fog density (%) | Adversary | $N_{\text{Exp-1}}$ | $N_{\text{Exp-2}}$ | $N$ | Detection before $sd$ | $d_{\mathcal{S}}$ (m) | Critical-zone FNR |
|---|---|---|---|---|---|---|---|
| 0.00 | False | 9 | 48 | 57 | 45.6% [33.3%, 57.9%] | 22.55 [13.55, 25.55] | 0.84 [0.79, 0.89] |
| 0.00 | True | 27 | 18 | 45 | 2.2% [0.0%, 6.7%] | 19.55 [13.55, 19.55] | 0.99 [0.98, 1.00] |
| 14.29 | False | 0 | 48 | 48 | 41.7% [27.1%, 56.2%] | 19.55 [16.55, 26.30] | 0.81 [0.73, 0.88] |
| 14.29 | True | 0 | 18 | 18 | 0.0% [0.0%, 0.0%] | 18.05 [13.55, 22.55] | 1.00 [1.00, 1.00] |
| 28.57 | False | 0 | 48 | 48 | 35.4% [22.9%, 50.0%] | 19.55 [15.80, 25.55] | 0.86 [0.81, 0.92] |
| 28.57 | True | 0 | 18 | 18 | 0.0% [0.0%, 0.0%] | 16.55 [13.55, 19.55] | 1.00 [1.00, 1.00] |
| 33.33 | False | 9 | 0 | 9 | 66.7% [33.3%, 88.9%] | 25.55 [22.55, 25.55] | 0.81 [0.69, 0.92] |
| 33.33 | True | 27 | 0 | 27 | 3.7% [0.0%, 11.1%] | 16.55 [12.20, 18.05] | 0.99 [0.97, 1.00] |
| 42.86 | False | 0 | 48 | 48 | 25.0% [14.6%, 37.5%] | 19.55 [13.55, 23.53] | 0.94 [0.91, 0.96] |
| 42.86 | True | 0 | 18 | 18 | 0.0% [0.0%, 0.0%] | 16.55 [13.55, 16.78] | 1.00 [1.00, 1.00] |
| 57.14 | False | 0 | 48 | 48 | 0.0% [0.0%, 0.0%] | 16.55 [13.55, 22.55] | 1.00 [1.00, 1.00] |
| 57.14 | True | 0 | 18 | 18 | 0.0% [0.0%, 0.0%] | 16.55 [13.55, 19.55] | 1.00 [1.00, 1.00] |
| 66.67 | False | 9 | 0 | 9 | 0.0% [0.0%, 0.0%] | 22.55 [22.55, 22.55] | 1.00 [1.00, 1.00] |
| 66.67 | True | 27 | 0 | 27 | 0.0% [0.0%, 0.0%] | 16.55 [10.55, 19.55] | 1.00 [1.00, 1.00] |
| 71.43 | False | 0 | 48 | 48 | 4.2% [0.0%, 10.4%] | 15.05 [13.55, 22.55] | 0.99 [0.97, 1.00] |
| 71.43 | True | 0 | 18 | 18 | 0.0% [0.0%, 0.0%] | 15.20 [10.55, 16.55] | 1.00 [1.00, 1.00] |
| 85.71 | False | 0 | 48 | 48 | 14.6% [6.2%, 25.0%] | 13.55 [13.55, 19.55] | 0.96 [0.94, 0.98] |
| 85.71 | True | 0 | 19 | 19 | 0.0% [0.0%, 0.0%] | 13.55 [10.55, 16.55] | 1.00 [1.00, 1.00] |
| 100.00 | False | 0 | 48 | 48 | 4.2% [0.0%, 10.4%] | 13.55 [13.55, 16.55] | 0.99 [0.97, 1.00] |
| 100.00 | True | 0 | 17 | 17 | 0.0% [0.0%, 0.0%] | 13.55 [10.55, 13.55] | 1.00 [1.00, 1.00] |

Table 3: Cloudiness sensitivity with precipitation and fog density fixed to zero, reported separately for adversary-inactive and adversary-active scenarios. A scenario is counted as successful if the calibrated ensemble mean reaches $\theta_{\mathcal{S}} = 0.75$ before the stopping-distance boundary. Detection-before-stopping values are reported as percentages with 95% bootstrap confidence intervals in brackets. $d_{\mathcal{S}}$ is reported as median [IQR]. Critical-zone FNR is computed over frames whose distance to the stop sign lies in $[sd, sd + \Delta]$, where $\Delta = 10$ m.

| Cloudiness (%) | Adversary | $N_{\text{Exp-1}}$ | $N_{\text{Exp-2}}$ | $N$ | Detection before $sd$ | $d_{\mathcal{S}}$ (m) | Critical-zone FNR |
|---|---|---|---|---|---|---|---|
| 0.00 | False | 9 | 48 | 57 | 45.6% [33.3%, 57.9%] | 22.55 [13.55, 25.55] | 0.84 [0.79, 0.89] |
| 0.00 | True | 27 | 18 | 45 | 2.2% [0.0%, 6.7%] | 19.55 [13.55, 19.55] | 0.99 [0.98, 1.00] |
| 33.33 | False | 9 | 0 | 9 | 88.9% [66.7%, 100.0%] | 25.55 [25.55, 28.55] | 0.69 [0.58, 0.81] |
| 33.33 | True | 27 | 0 | 27 | 3.7% [0.0%, 11.1%] | 19.55 [13.55, 22.55] | 0.99 [0.97, 1.00] |
| 66.67 | False | 9 | 0 | 9 | 100.0% [100.0%, 100.0%] | 25.55 [25.55, 25.55] | 0.72 [0.67, 0.75] |
| 66.67 | True | 27 | 0 | 27 | 0.0% [0.0%, 0.0%] | 19.55 [13.55, 22.55] | 1.00 [1.00, 1.00] |

Table 4: Precipitation sensitivity with cloudiness and fog density fixed to zero, reported separately for adversary-inactive and adversary-active scenarios. A scenario is counted as successful if the calibrated ensemble mean reaches $\theta_{\mathcal{S}} = 0.75$ before the stopping-distance boundary. Detection-before-stopping values are reported as percentages with 95% bootstrap confidence intervals in brackets. $d_{\mathcal{S}}$ is reported as median [IQR]. Critical-zone FNR is computed over frames whose distance to the stop sign lies in $[sd, sd + \Delta]$, where $\Delta = 10$ m.

| Precipitation (%) | Adversary | $N_{\text{Exp-1}}$ | $N_{\text{Exp-2}}$ | $N$ | Detection before $sd$ | $d_{\mathcal{S}}$ (m) | Critical-zone FNR |
|---|---|---|---|---|---|---|---|
| 0.00 | False | 9 | 48 | 57 | 45.6% [33.3%, 57.9%] | 22.55 [13.55, 25.55] | 0.84 [0.79, 0.89] |
| 0.00 | True | 27 | 18 | 45 | 2.2% [0.0%, 6.7%] | 19.55 [13.55, 19.55] | 0.99 [0.98, 1.00] |
| 33.33 | False | 9 | 0 | 9 | 88.9% [66.7%, 100.0%] | 28.55 [25.55, 28.55] | 0.64 [0.53, 0.75] |
| 33.33 | True | 27 | 0 | 27 | 3.7% [0.0%, 11.1%] | 19.55 [13.55, 22.55] | 0.99 [0.97, 1.00] |
| 66.67 | False | 9 | 0 | 9 | 88.9% [66.7%, 100.0%] | 25.55 [25.55, 28.55] | 0.69 [0.61, 0.81] |
| 66.67 | True | 27 | 0 | 27 | 3.7% [0.0%, 11.1%] | 19.55 [13.55, 22.55] | 0.99 [0.97, 1.00] |

# 3 Calibration

Table 5: COCO-val stop-sign confidence calibration results.

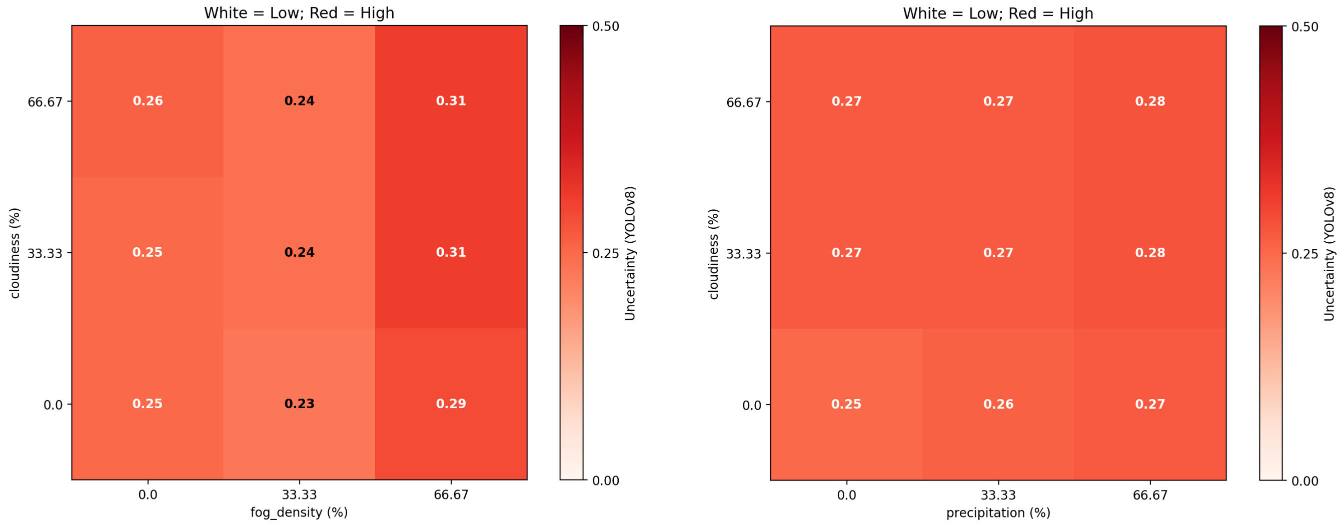
| Model | $N$ | Correct | Raw ECE | Cal. ECE | Raw NLL | Cal. NLL |
|---|---|---|---|---|---|---|
| YOLOv8 | 1000 | 72 | 0.0133 | 0.0124 | 0.0947 | 0.0926 |
| YOLOv9 | 737 | 71 | 0.0197 | 0.0132 | 0.1164 | 0.1130 |
| RT-DETR | 5756 | 68 | 0.0328 | 0.0030 | 0.0445 | 0.0163 |
| DETR50 | 776 | 65 | 0.0411 | 0.0158 | 0.1314 | 0.0833 |
| DETR101 | 901 | 64 | 0.0366 | 0.0121 | 0.1157 | 0.0783 |

## 4 Real-world Test Vectors

Table 6: Real-world validation test vectors. Each test vector corresponds to a camera-based stop-sign driving sequence collected under a distinct combination of speed, lighting, weather, and occlusion condition at Virginia Tech Transportation Institute (VTTI) Testing Facility [**?**].

| **N** | **Speed** (mph) | **Light** | **Weather** | **Occlusion** (Pedestrian, Cars, Vegetation) | Replication |
|---|---|---|---|---|---|
| 1 | 25 | Day | No rain | No occlusion | 3 |
| 2 | 25 | Day | No rain | Occlusion | 3 |
| 3 | 25 | Day | Artificial Rain | No occlusion | 3 |
| 4 | 25 | Day | Artificial Rain | Occlusion | 3 |
| 5 | 25 | Night | No rain | No occlusion | 3 |
| 6 | 25 | Night | No rain | Occlusion | 3 |
| 7 | 25 | Night | Artificial Rain | No occlusion | 3 |
| 8 | 25 | Night | Artificial Rain | Occlusion | 3 |
| 9 | 35 | Day | No rain | No occlusion | 3 |
| 10 | 35 | Day | No rain | Occlusion | 3 |
| 11 | 35 | Day | Artificial Rain | No occlusion | 3 |
| 12 | 35 | Day | Artificial Rain | Occlusion | 3 |
| 13 | 35 | Night | No rain | No occlusion | 3 |
| 14 | 35 | Night | No rain | Occlusion | 3 |
| 15 | 35 | Night | Artificial Rain | No occlusion | 3 |
| 16 | 35 | Night | Artificial Rain | Occlusion | 3 |
| 17 | 55 | Day | No rain | No occlusion | 3 |
| 18 | 55 | Day | No rain | Occlusion | 3 |
| 19 | 55 | Day | Artificial Rain | No occlusion | 3 |
| 20 | 55 | Day | Artificial Rain | Occlusion | 3 |
| 21 | 55 | Night | No rain | No occlusion | 3 |
| 22 | 55 | Night | No rain | Occlusion | 3 |
| 23 | 55 | Night | Artificial Rain | No occlusion | 3 |
| 24 | 55 | Night | Artificial Rain | Occlusion | 3 |

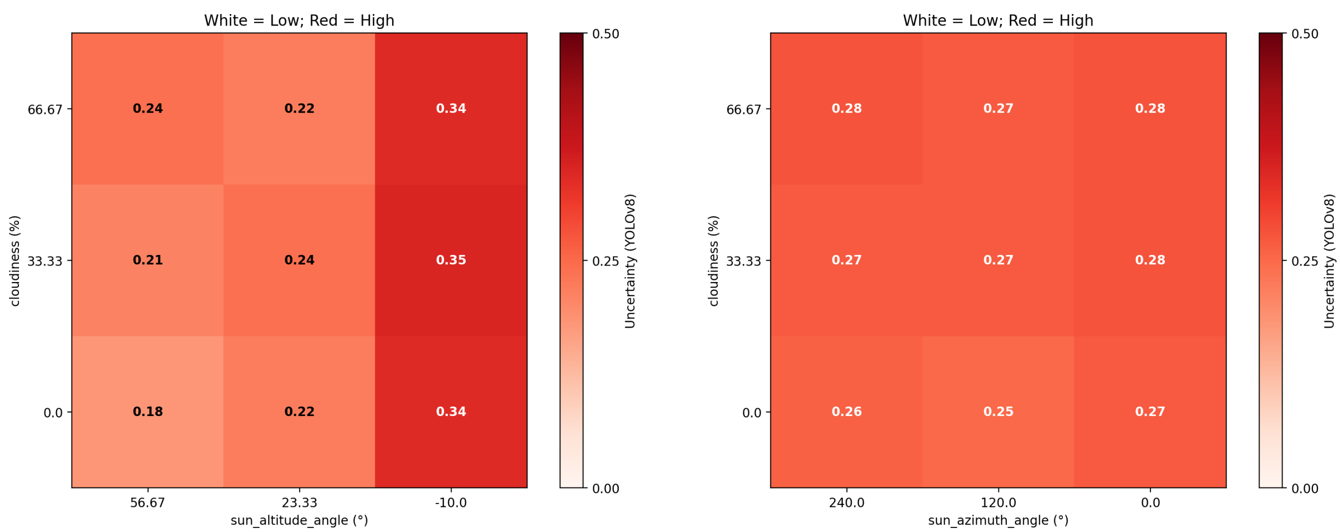
## 5 Comparative Heatmap and Ensemble plots for Real-World VTTI Samples




(a) Sensitivity: Cloudiness vs Fog density

(b) Sensitivity: Cloudiness vs Precipitation

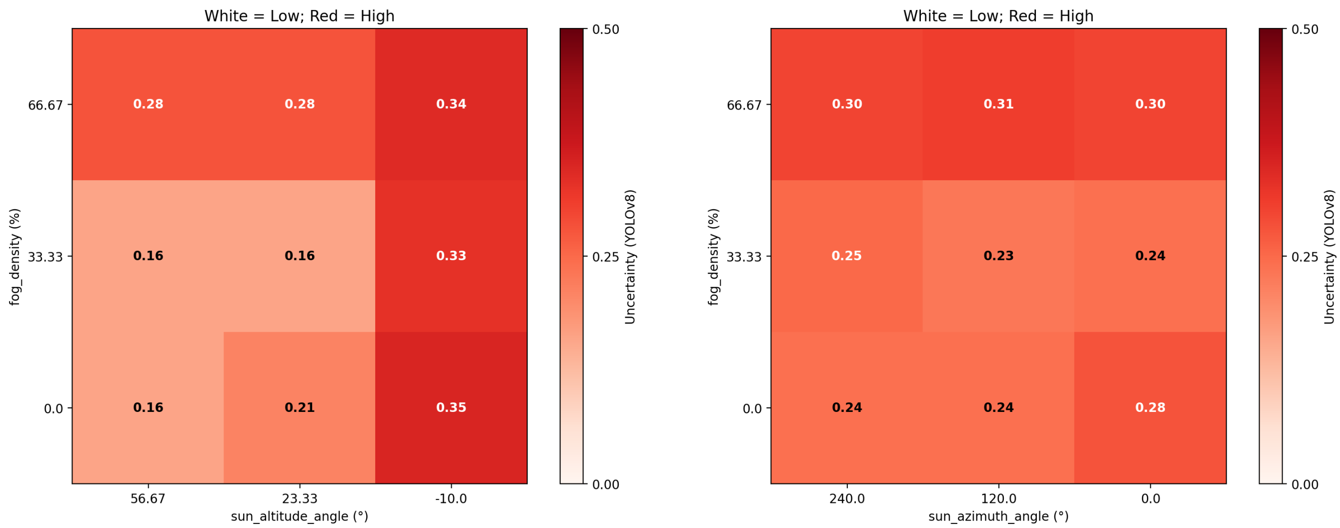
Figure 1: Sensitivity analysis of Cloudiness vs Fog density and Cloudiness vs Precipitation




(a) Sensitivity: Cloudiness vs Sun Altitude Angle

(b) Sensitivity: Cloudiness vs Sun Azimuth Angle

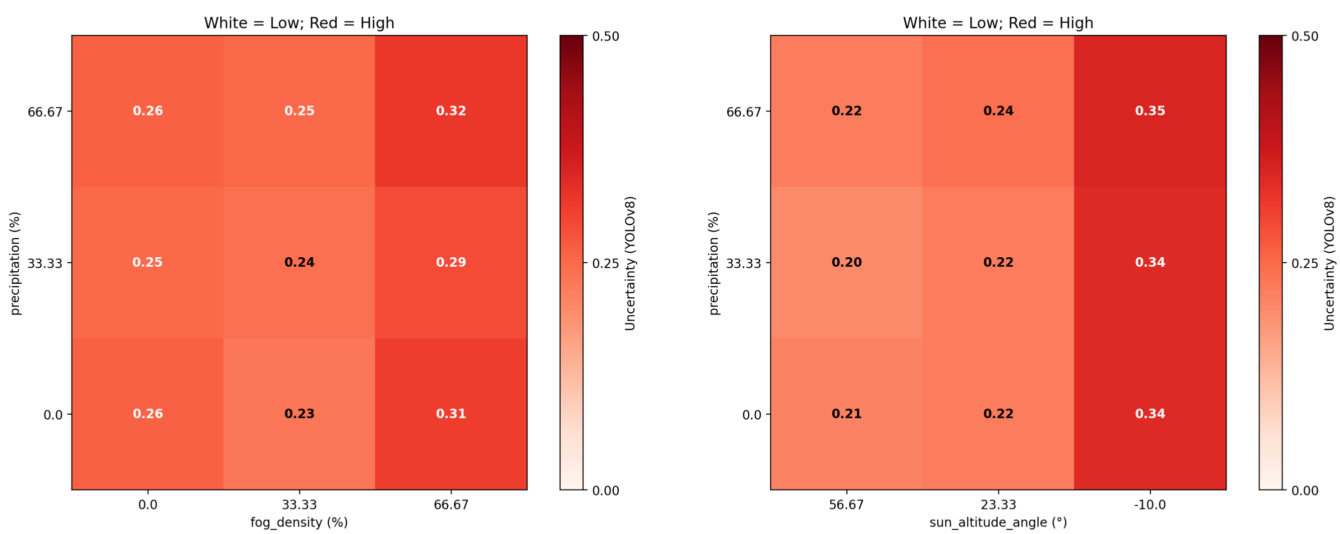
Figure 2: Sensitivity analysis of Cloudiness vs Sun Altitude Angle and Cloudiness vs Sun Azimuth Angle




(a) Sensitivity: Fog Density vs Sun Altitude Angle

(b) Sensitivity: Fog Density vs Sun Azimuth Angle

Figure 3: Sensitivity analysis of Fog Density vs Sun Altitude Angle and Fog Density vs Sun Azimuth Angle




(a) Sensitivity: Precipitation vs Fog Density

(b) Sensitivity: Precipitation vs Sun Altitude Angle

Figure 4: Sensitivity analysis of Precipitation vs Fog Density and Precipitation vs Sun Altitude Angle

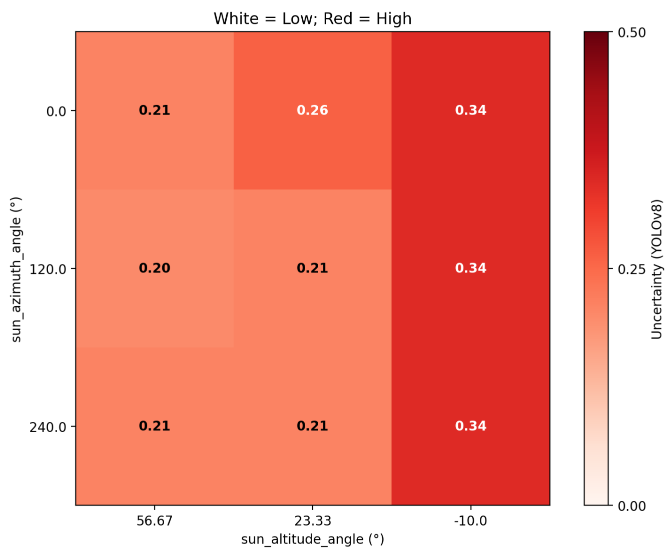
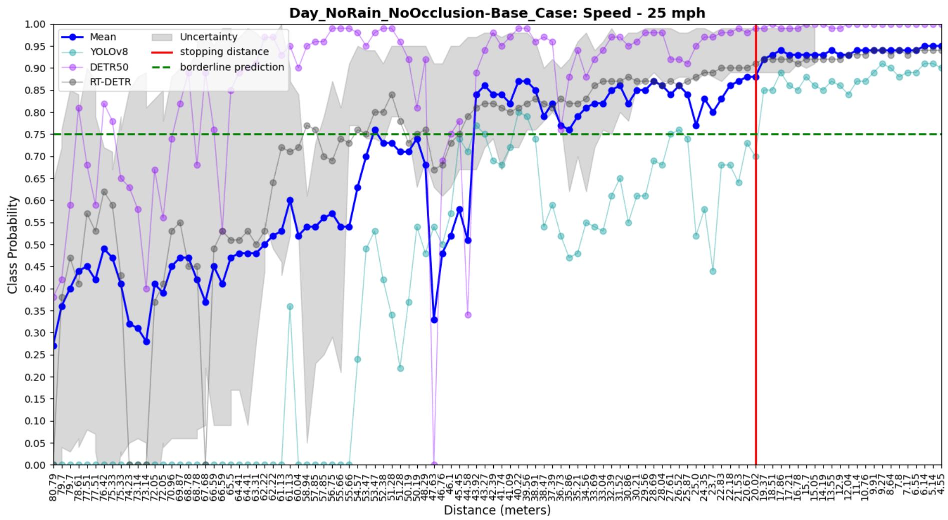
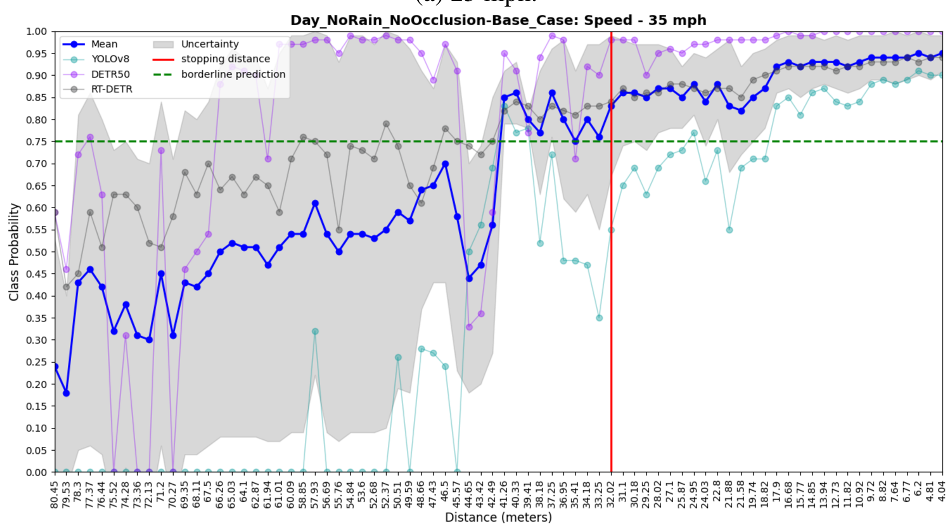


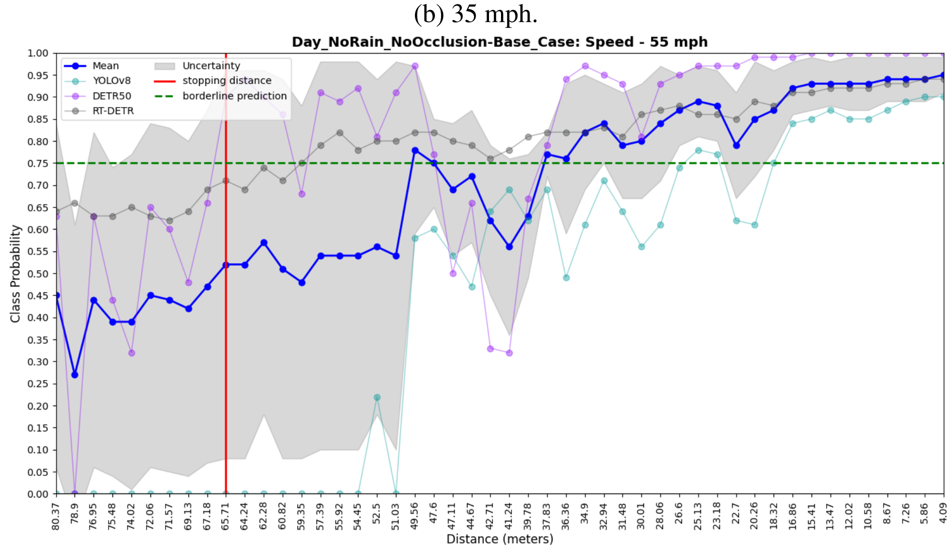

Figure 5: Sensitivity: Sun Altitude Angle vs Sun Azimuth Angle




(a) 25 mph.

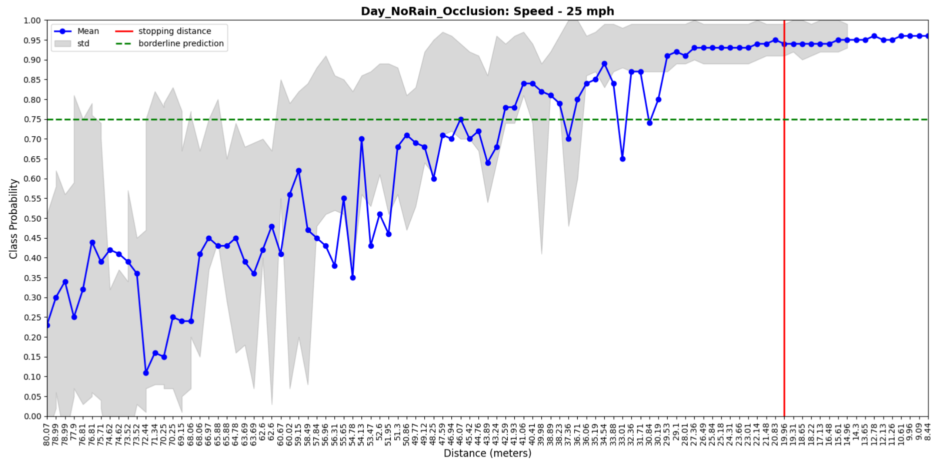



(b) 35 mph.

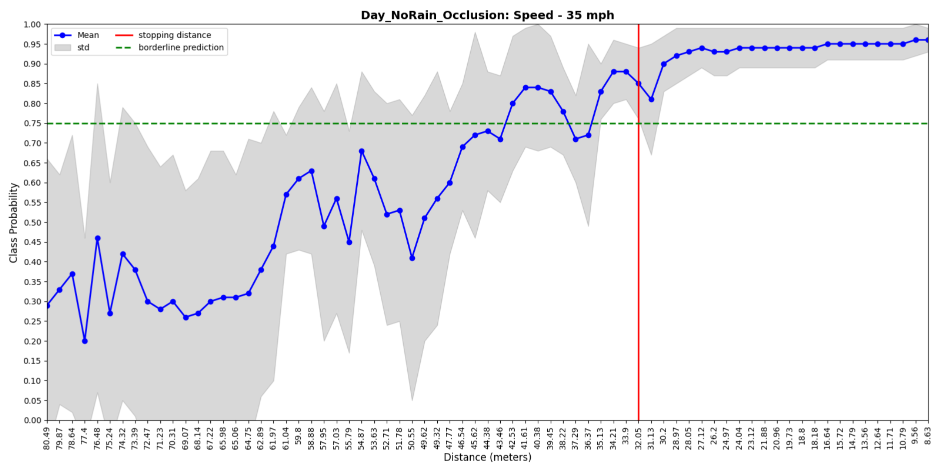



(c) 55 mph.

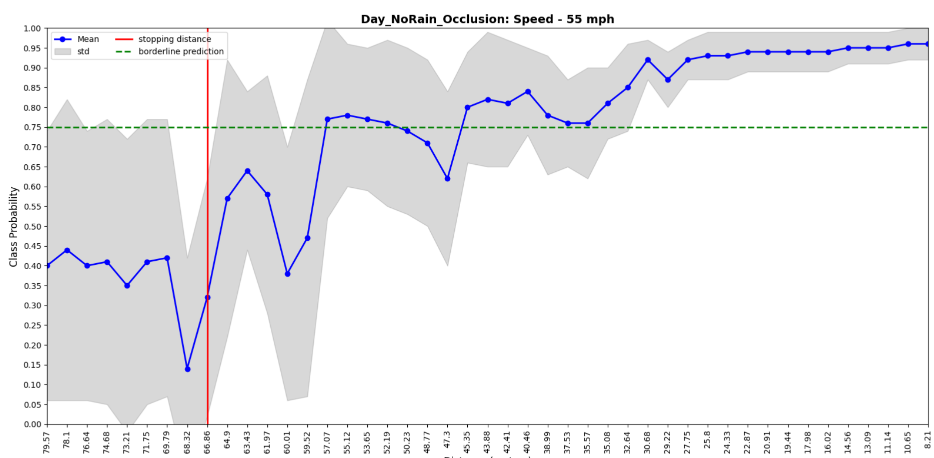
Figure 6: Day_NoRain_NoOcclusion-Base_Case




(a) 25 mph.

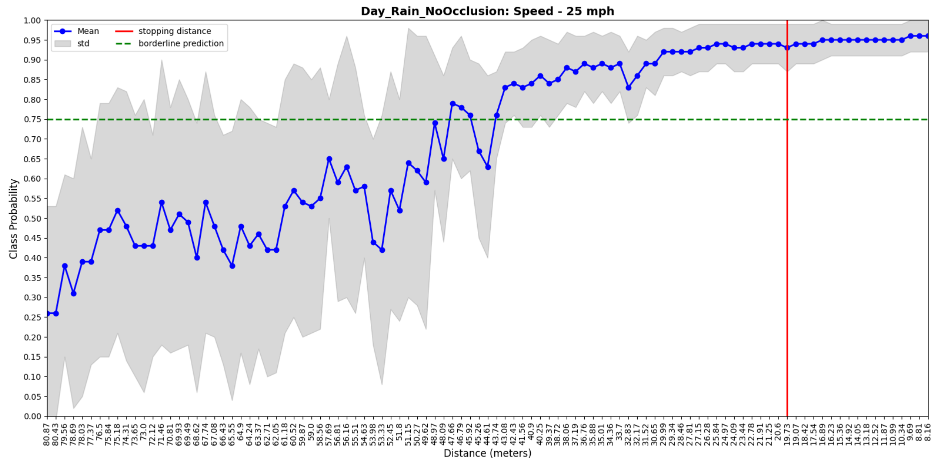



(b) 35 mph.

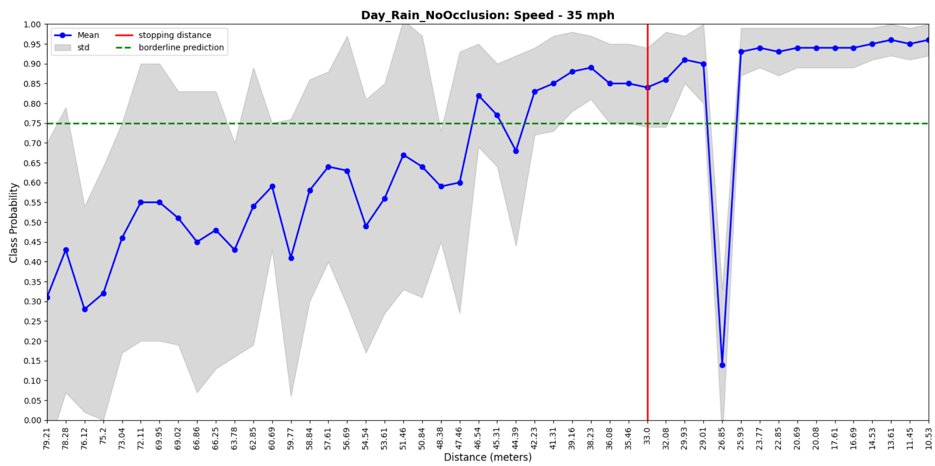



(c) 55 mph.

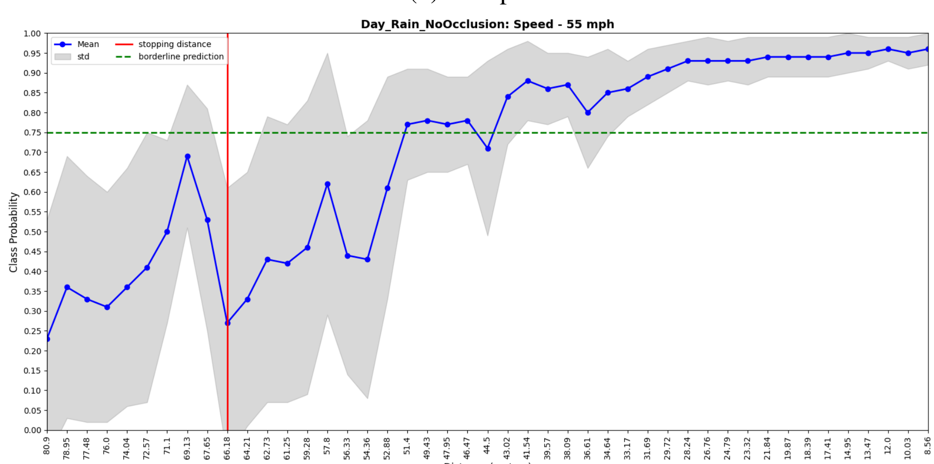
Figure 7: Day_NoRain_Occlusion




(a) 25 mph.

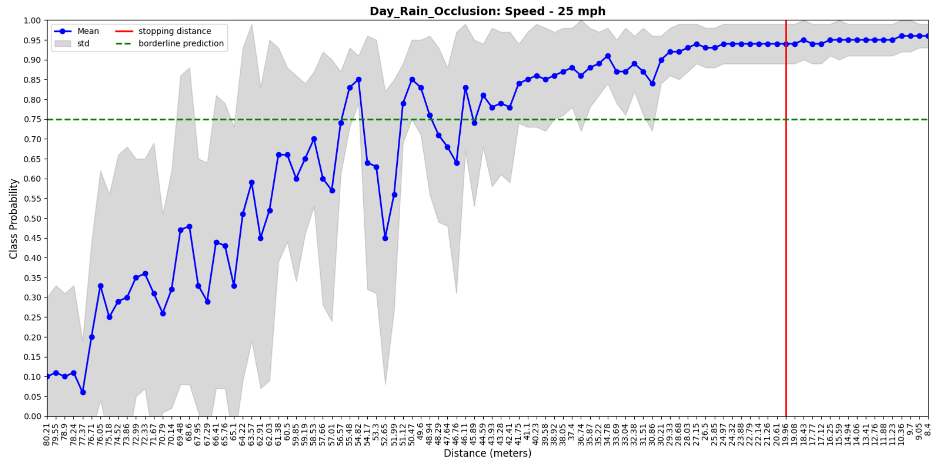



(b) 35 mph.

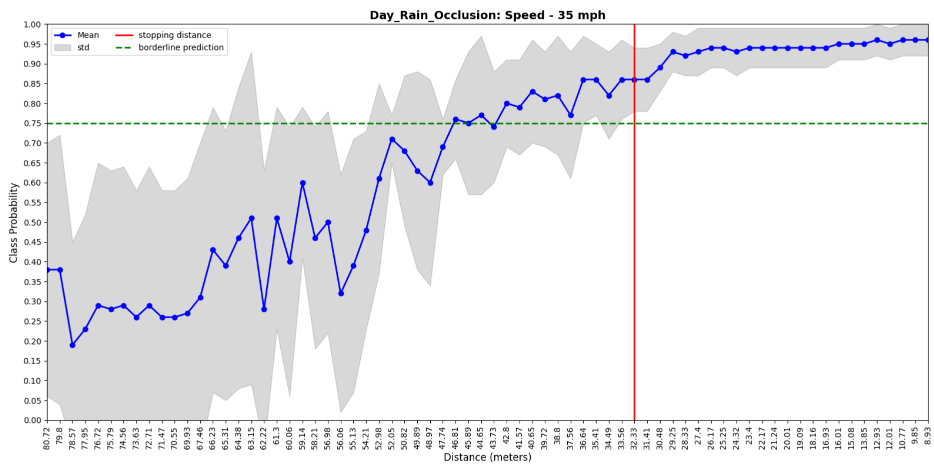



(c) 55 mph.

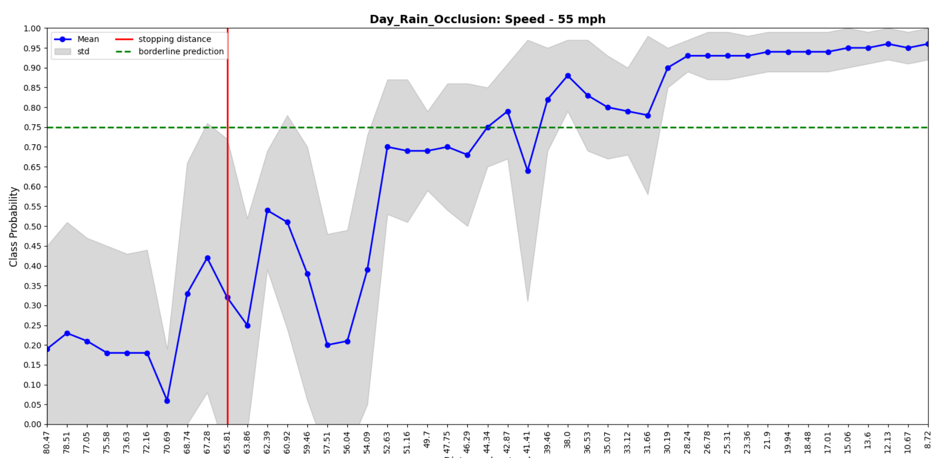
Figure 8: Day_Rain_NoOcclusion




(a) 25 mph.

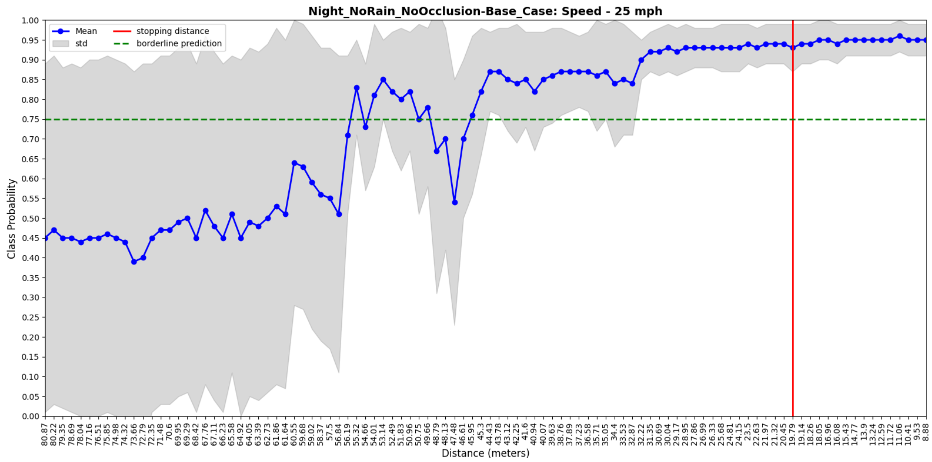



(b) 35 mph.

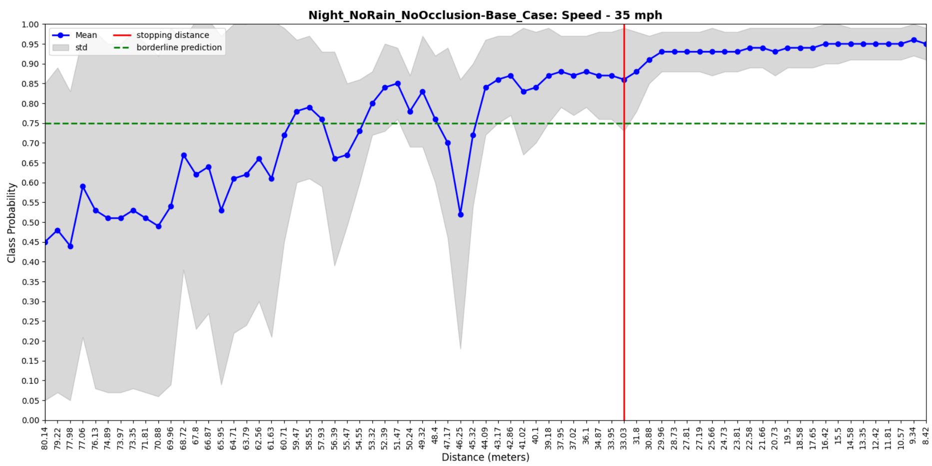



(c) 55 mph.

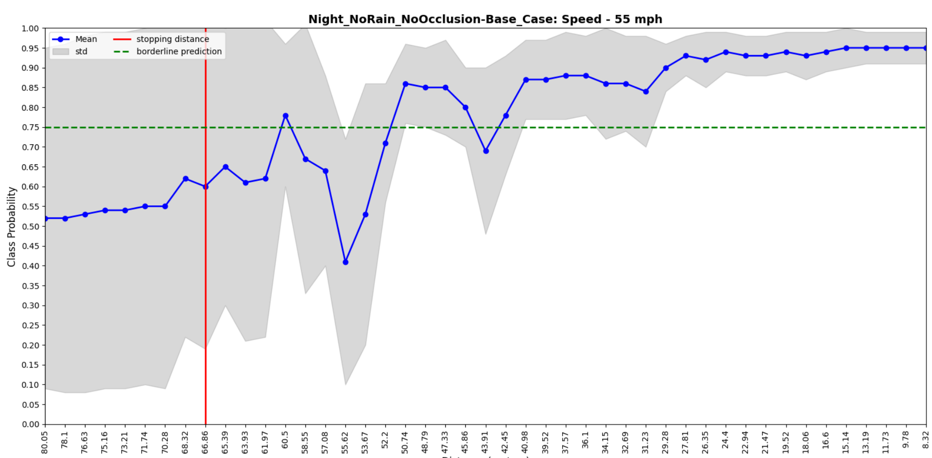
Figure 9: Day_Rain_Occlusion




(a) 25 mph.

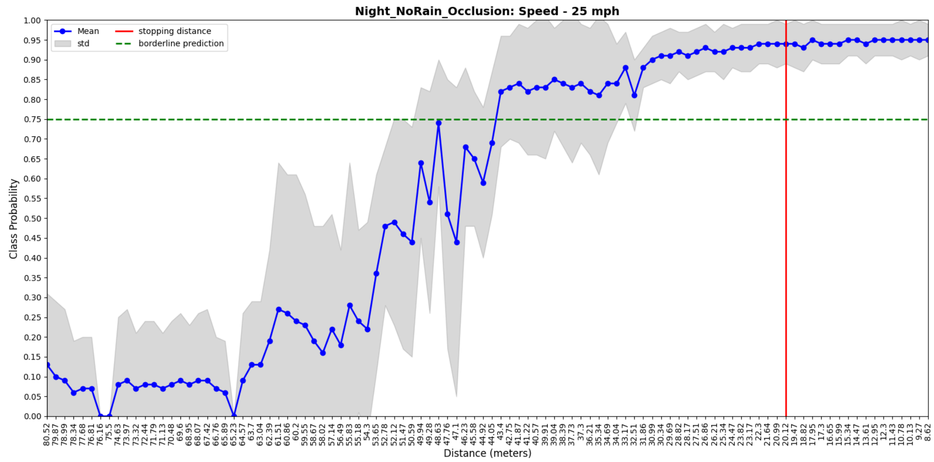



(b) 35 mph.

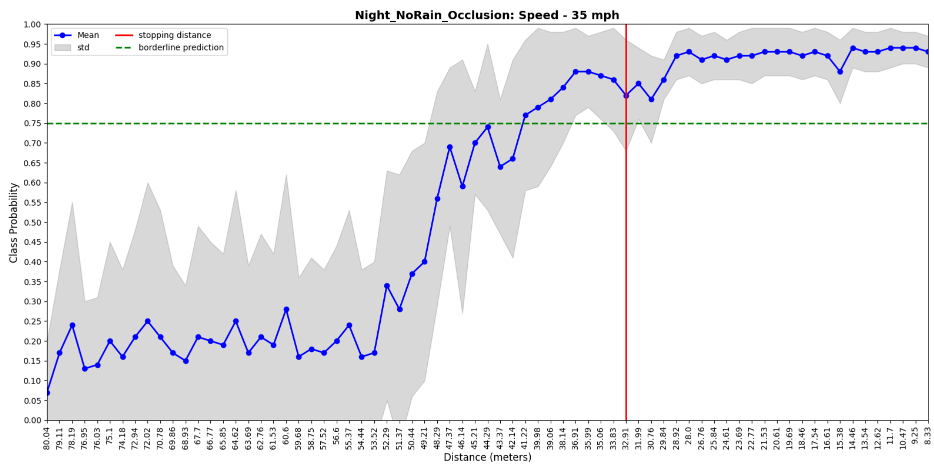



(c) 55 mph.

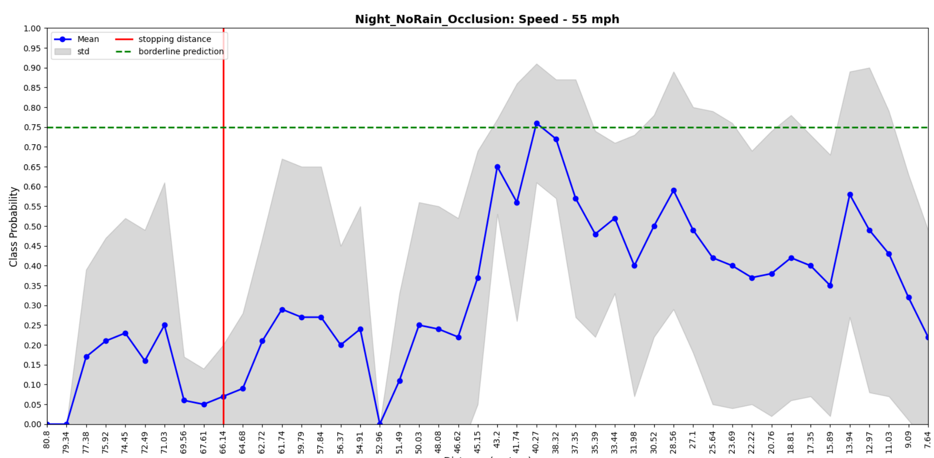
Figure 10: Night_NoRain_NoOcclusion-Base_Case




(a) 25 mph.

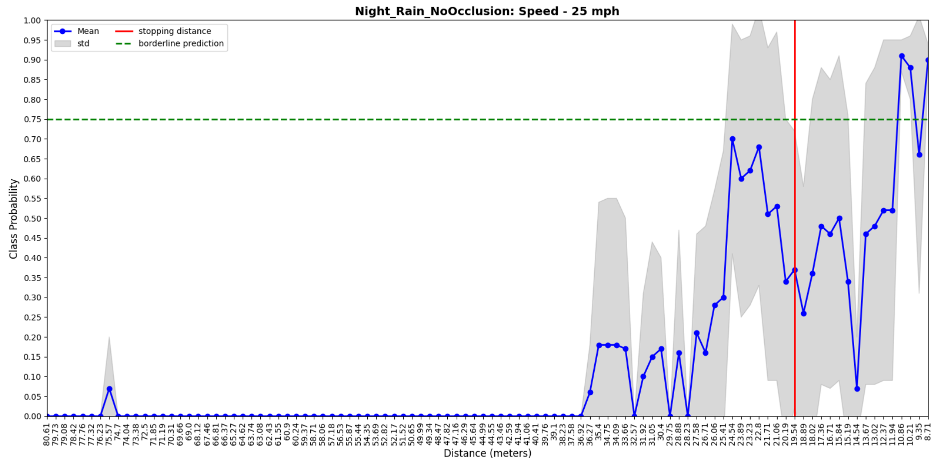



(b) 35 mph.

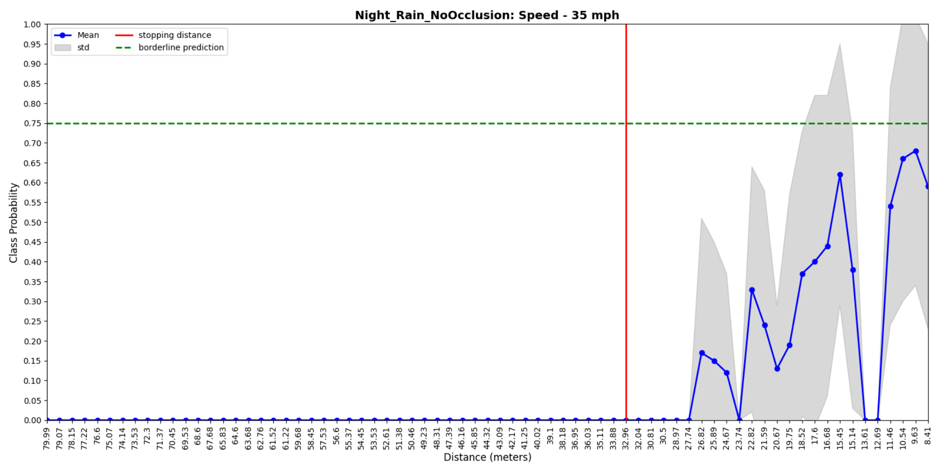



(c) 55 mph.

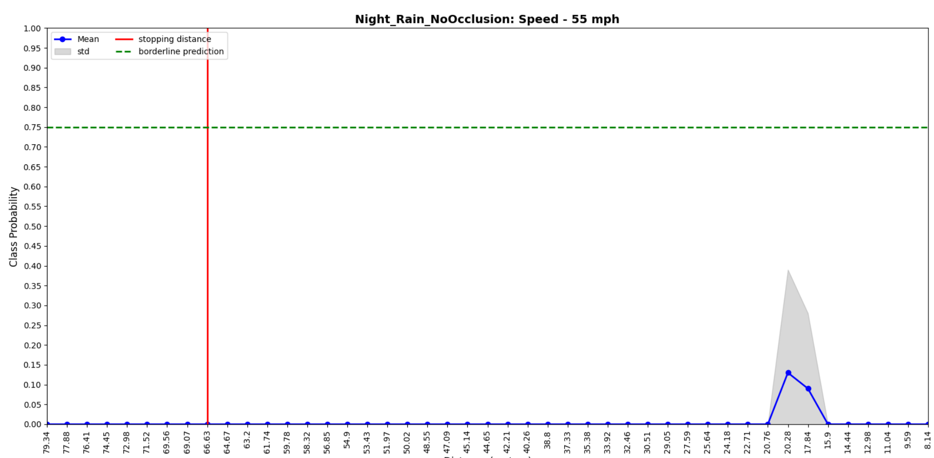
Figure 11: Night_NoRain_Occlusion




(a) 25 mph.

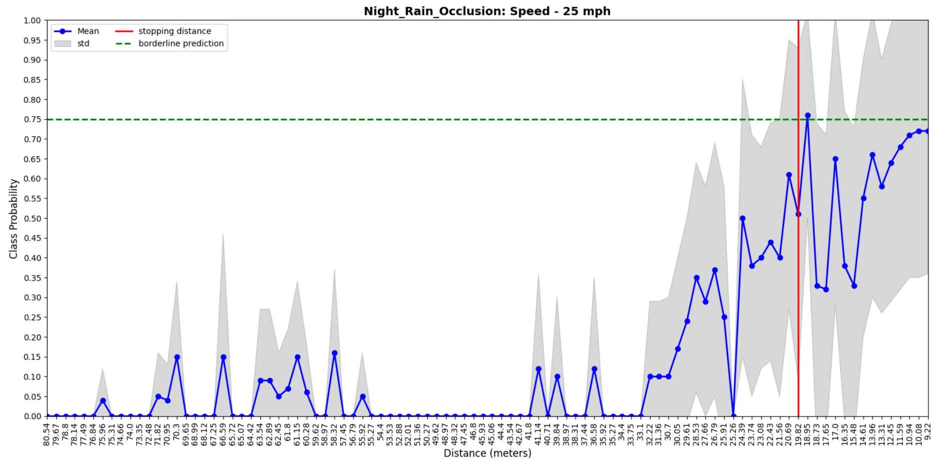



(b) 35 mph.

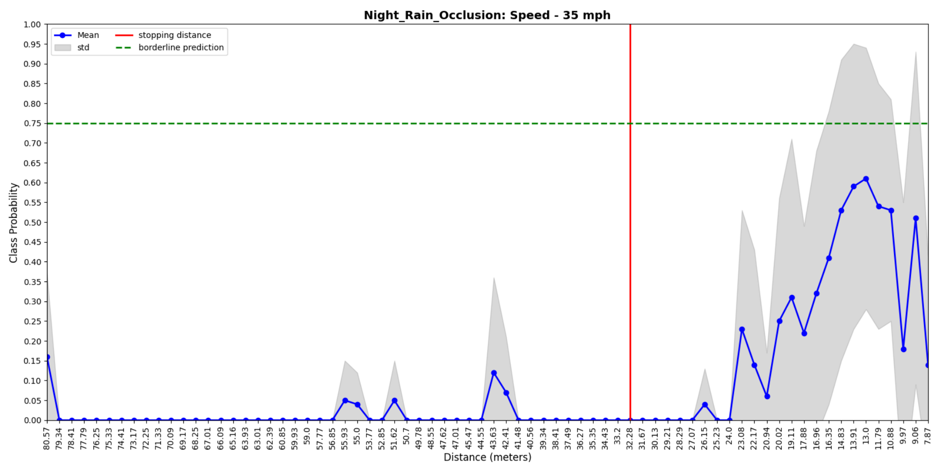



(c) 55 mph.

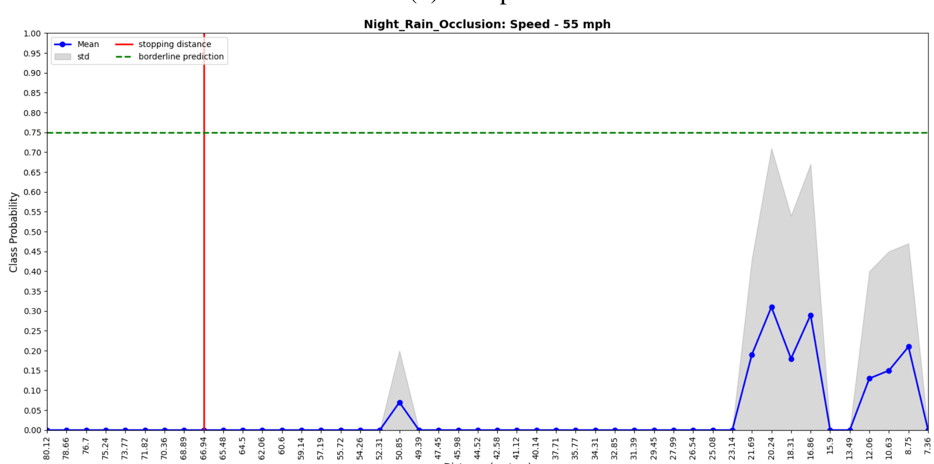
Figure 12: Night_Rain_NoOcclusion




(a) 25 mph.




(b) 35 mph.




(c) 55 mph.

Figure 13: Night_Rain_Occlusion